

CSE5DMI
Data Mining

Week 12

La Trobe University CRICOS Provider Code Number 00115M

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STUDENT FEEDBACK ON SUBJECTS AND TEACHING



STUDENT FEEDBACK ON SUBJECTS AND TEACHING

Your feedback is very important to La Trobe. Student feedback is used by teaching staff to improve the quality of subjects and the student experience.

Previous student feedback has made a positive impact on how subjects are delivered.

You will have received an email invitation from surveys@latrobe.edu.au to complete your subject and teacher evaluations.

Devices e.g. laptops, smartphones and tablets can be used to complete your evaluations.

Students are expected to communicate in a respectful, safe and tolerant manner.

All survey responses are kept confidential. However, the University reserves the right to investigate inappropriate communications and identify person/s engaging in the provision of disrespectful, unsafe or intolerant feedback.

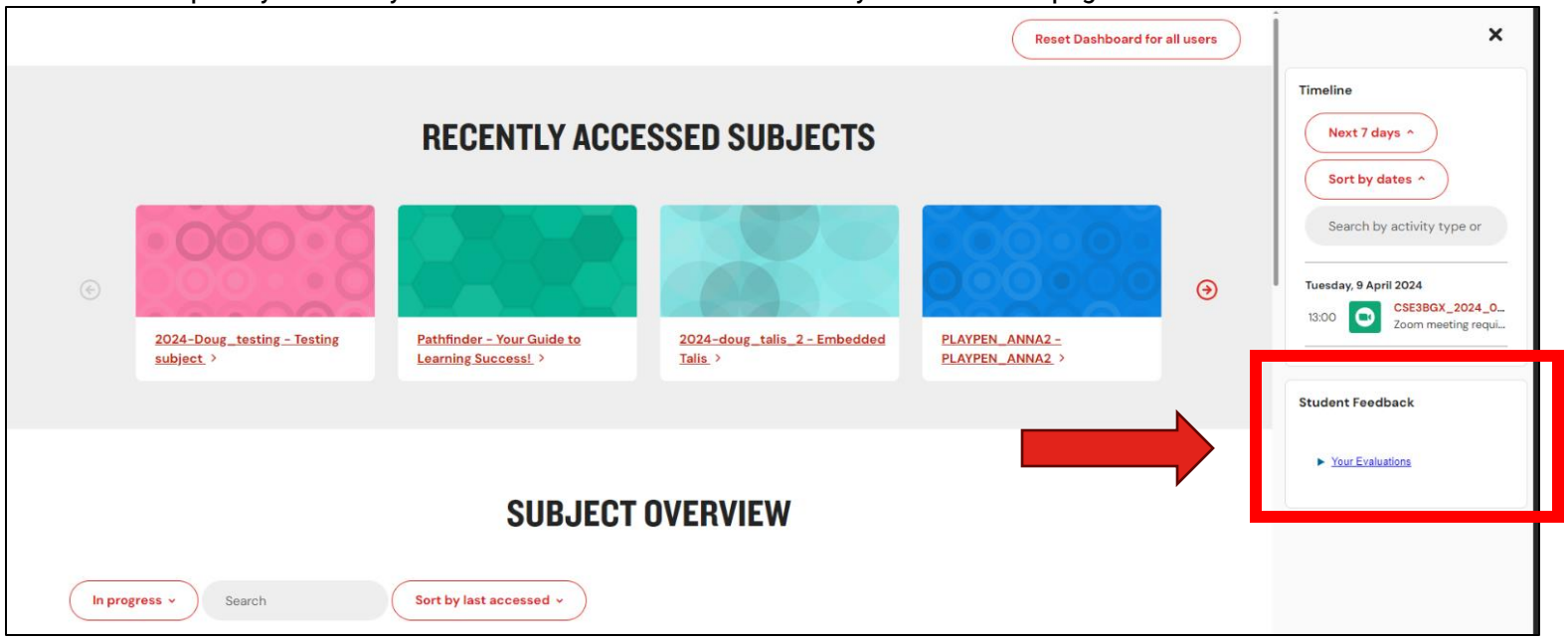
If you have any questions, please email surveys@latrobe.edu.au

HOW TO COMPLETE FEEDBACK

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You can also complete your surveys via the Student feedback block on your LMS homepage.



Week 12

- Final exam
- Summary and revision

- Assignment 2 consultation session #2: Tuesday1:00 pm to 3:00 pm,
- Online: Friday 11 am -12 pm

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Final Exam

- Date: 16 October 2026, 14:00
- Venue: Union Hall
- Time limit: 2 hours
- Reading time: 15 minutes
- Allowable materials
  - Non-programmable calculator
  - One A4 page of handwritten notes, 2 sides – no photocopies

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La Trobe University
Final Assessment Period Semester 2, 2026
RESTRICTED USE

This paper MUST NOT BE REMOVED from the examination venue

Student ID: ☐☐☐☐☐☐☐☐ Seat Number: ☐☐☐

Subject code:	CSE5DMI	Paper number:	1
Reading time:	15	minutes	
Writing time:	120	minutes	
Number of pages:	16	(including cover sheet)	

Campus:

☐ Albury-Wodonga ☐ Bendigo ☒ Bundoora ☐ City (Melbourne)
☐ Mildura ☐ Shepparton ☐ Sydney

Allowable materials:

Number:	02	Description:	Non-programmable calculator permitted
Number:	06	Description:	One A4 page of handwritten notes, 2 sided - no photocopies
Number:		Description:	
Number:		Description:	

Instructions to candidates:

1. Please write ALL your answers to the questions (including multiple choice questions) in the blank boxes provided in the exam paper.
2. Write your answers in blue or black pen, not pencil – if we cannot see or read your response clearly, we cannot give you a mark.
3. The exam contains 6 questions, worth a total of 120 marks (40% of subject marks).

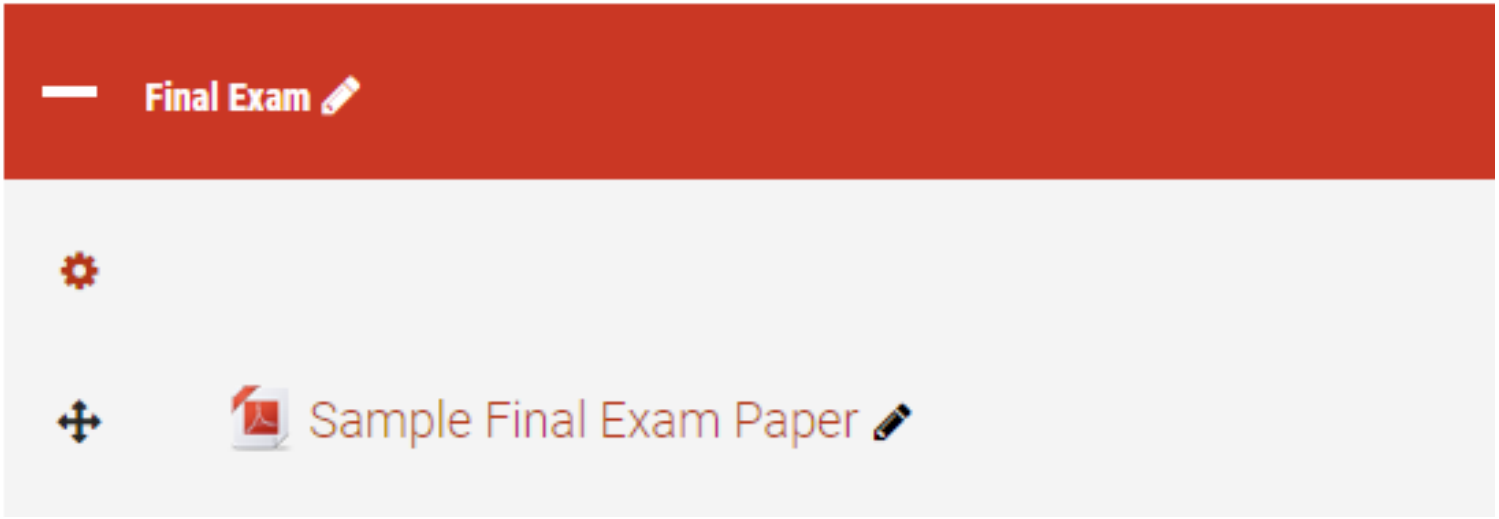
Exam questions

6 questions

- Question 1 [40 marks]: 6 multiple-choice questions
- Question 2 [10 marks]: calculation
- Question 3 [15 marks]: calculation
- Question 4 [15 marks]: calculation
- Question 5 [20 marks]: calculation
- Question 6 [20 marks]: calculation

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Sample Exam Questions



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LEARNING ACTIVITIES

Session	Learning Focus and Activities
1	Lecture – Introduction to Data Mining
2	Lecture – Data and Similarity
3	Lecture – Ruled-based Classification Techniques (1)
4	Lecture – Rule-based Classification Techniques (2)
5	Lecture – Learning-based Classification Techniques (1)
6	Lecture – Learning-based Classification Techniques (2)
7	Lecture – Basics of Association Analysis
8	Lecture –Association Analysis
9	Lecture –Rule-clustering Techniques (1)
10	Lecture –Rule-clustering Techniques (2)
11	Lecture –Rule-clustering Techniques (3)
12	Lecture – Review;
	Total

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Upon completion of this subject, you should be able to:

1. Explain the technologies and applications of data mining techniques (DM).
2. Explain the major components and issues in data mining techniques, such as pattern discovery through classification systems, association analysis and clustering algorithms.
3. Evaluate data mining algorithms, i.e. classification systems, important issues in clustering algorithms, and the links between data mining and knowledge engineering.
4. Implement data mining algorithms for computational discovery.

DATA MINING TECHNIQUES

- It plays a key role for building computational intelligence systems or expert systems that help in many data-driven applications
- It aims to extract potentially useful and interesting patterns from collected data
- It is the computational process of discovering patterns in big data sets
- Data mining techniques are involving methods at the intersection of machine learning, database systems, artificial intelligence, and statistics.

PART 1: CONCEPTS (LECTURES 1, 2 & 3)



1. What is data mining? Data mining tasks
2. Process of knowledge discovery
3. Definitions on classification, clustering and association rule mining
4. Application domains of data mining techniques
5. Challenges of data mining
6. Data set types
7. Data quality, outliers, missing values, duplicated data, etc.
8. Curse of dimensionality and dimensionality reduction
9. Data pre-processing techniques

PART 2: CLASSIFICATION TECHNIQUES (LECTURES 4, 5 & 6)

1. Over fitting and under fitting
2. Pre-pruning and post-pruning, optimistic and pessimistic errors
3. Calculate Gini index, entropy, information gain for given datasets
4. General description of Hunt's algorithm for building Decision Tree (DT), some issues in design
5. Confusion matrix and limits for classifiers' evaluation
6. Characteristics of rule-based classifier, inferring classification rules from decision tree, or vice versa
7. Metrics of rule-based classification systems (ROC curve, accuracy, cost, recall, precision, coverage rate, robustness)
8. Naïve Bayes classifier
9. What is SVM and ANN? Basic ideas behind them

PART 3: ASSOCIATION ANALYSIS (LECTURES 7 & 8)

1. Define association rule mining tasks
2. Basic concepts: itemset, frequent itemset, supports and confidence
3. Calculation of supports and confidence for a given dataset
4. Apriori principle: if an itemset is frequent, then all of its subsets must also be frequent
5. Description of Apriori Algorithm
6. Maximal vs closed itemsets
7. Interestingness evaluations of association rules, and their drawbacks

PART 4: CLUSTERING TECHNIQUES (LECTURES 9, 10 & 11)

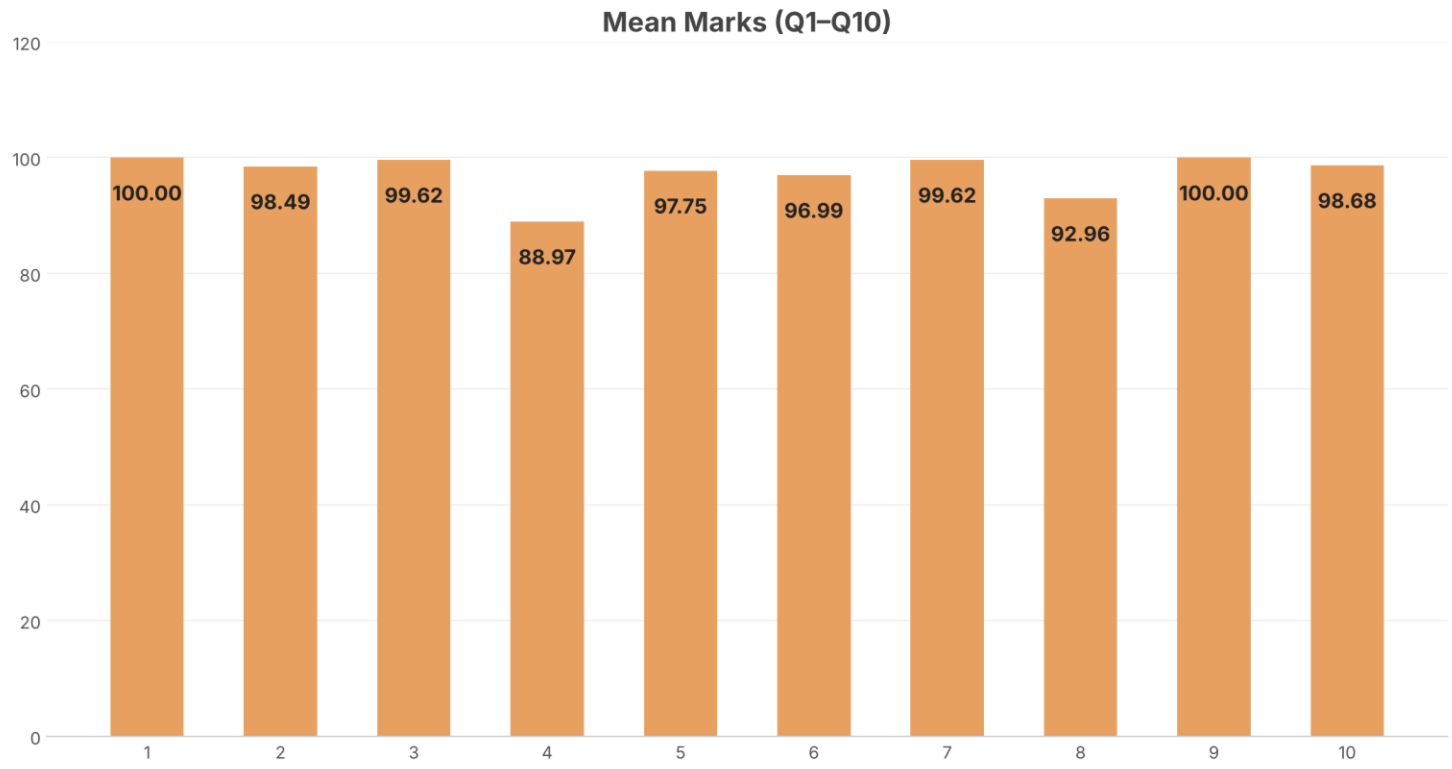
1. Description of the basic K-Means clustering algorithm, and evaluation of this clustering algorithm
2. Issues related to K-Means clustering techniques
3. Importance of choosing the initial centroids in K-Means clustering method
4. Hierarchical Clustering and proximity matrix
5. Basics of DBSCAN
6. Strengths and weaknesses of each clustering algorithm
7. How to measure the quality of clustering results?

REMARKS

- All contents taught in lectures are possibly appear in the exam papers
- Read the question carefully to get understanding before writing

Good luck!

Feedback on Quizzes



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Decision Tree

- How to learn a tree model
  - Basic elements in a tree
    - Tree nodes
    - Attribute test
  - How to split the records
    - How to specify the attribute test condition
    - How to find the best split

Nominal? Ordinal? Continuous?

- Depends on attribute type
- Depends on number of ways to split

2-way or multi-way?

Measure the homogeneity / purity
- Gini index
- Entropy

- prefer larger *GINI_**split***

= smaller child node Gini (given the same parent node)

- prefer larger *information_**gain***

= smaller child node entropy (given the same parent node)

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Model Evaluation

All the data we have	Training set	Validation set	Test set
Purpose	• Training process/ model building process		• Evaluation process
	• Avoid underfitting / overfitting • Select the best model		• Estimate the performance on unseen data
methods	• Early stopping/post-pruning		
	• estimating generalization error	• Metrics for performance evaluation (confusion matrix, etc)	

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Model Evaluation

- Optimistic error

= Error rate on *training set*

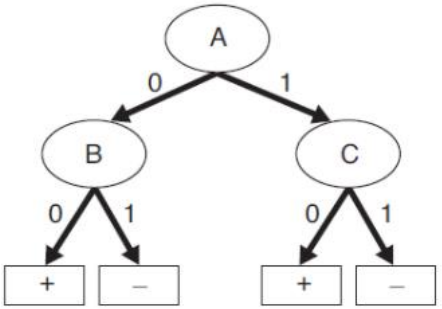
- Pessimistic error

calculation involves:
- Error rate on *training set*
- Trade-off hyper-parameter
- Number of leaf nodes
- Total number of training records

- Confusion matrix
- Cost matrix
- Accuracy, precision, recall F-measure

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Week 5 Quiz 4



Training:

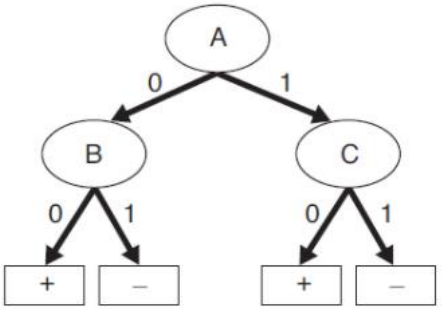
Instance	A	B	C	Class
1	0	0	0	+
2	0	0	1	+
3	0	1	0	+
4	0	1	1	–
5	1	0	0	+
6	1	0	0	+
7	1	1	0	–
8	1	0	1	+
9	1	1	0	–
10	1	1	0	–

Validation:

Instance	A	B	C	Class
11	0	0	0	+
12	0	1	1	+
13	1	1	0	+
14	1	0	1	–
15	1	0	0	+

- 1) Given the decision tree and datasets as above, the generalization error rate of the tree using the optimistic approach is (value only, 1 decimal point)
- 2) Given the decision tree and datasets as above, the generalization error rate of the tree using the pessimistic approach (simply use the strategy of adding a factor of 0.5 to each leaf node) is: (value only, 1 decimal point)
- 3) Given the decision tree and datasets as above, the accuracy of the tree using the validation set is: (value only, 1 decimal point)

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Training:

Instance	A	B	C	Class
1	0	0	0	+
2	0	0	1	+
3	0	1	0	+
4	0	1	1	–
5	1	0	0	+
6	1	0	0	+
7	1	1	0	–
8	1	0	1	+
9	1	1	0	–
10	1	1	0	–

Validation:

Instance	A	B	C	Class
11	0	0	0	+
12	0	1	1	+
13	1	1	0	+

Recall what we have learnt in Week 3

Key points:

- How to trace a given decision tree?
- Optimistic error and pessimistic errors
- Confusion matrix

Recall what we have learnt in Week 4

All the data we have	Training set	Validation set	Test set
Purpose	• Training process/ model building process		• Evaluation process
	• Avoid underfitting / overfitting • Select the best model		• Estimate the performance on unseen data
methods	• Early stopping/post-pruning		
	• estimating generalization error		• Metrics for performance evaluation (confusion matrix, etc)

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Week 6

Given the dataset in the table

Record	A	B	C	Class
1	0	0	0	+
2	0	0	1	-
3	0	1	1	-
4	0	1	1	-
5	0	0	1	+
6	1	0	1	+
7	1	0	1	-
8	1	0	1	-
9	1	1	1	+
10	1	0	1	+

estimate the conditional probabilities for P(A=0 | -) and fill in the blank. (Value only, keep 1 decimal point)

estimate the probability for P(-) and fill in the blank. (Value only, keep 1 decimal point)

Let's use the Bayes classifier we have introduced in the lecture to predict the class label for a test sample. Given the attribute values of a test sample as (A=0, B=1, C=0), what is the class label of this sample?

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Given the dataset in the table

Record	A	B	C	Class
1	0	0	0	+
2	0	0	1	-
3	0	1	1	-
4	0	1	1	-
5	0	0	1	+
6	1	0	1	+
7	1	0	1	-
8	1	0	1	-
9	1	1	1	+
10	1	0	1	+

estimate the conditional probabilities for P(A=0 | -) and fill in the blank. (V

estimate the probability for P(-) and fill in the blank. (Value only, keep 1 d

Let's use the Bayes classifier we have introduced in the lec
values of a test sample as (A=0, B=1, C=0), what is the cla

- P(A=0|class=-)=3/5
- Class:
  - P(class=-)=5/10
  - P(class=+)=5/10
- P(-|A=0, B=1, C=0)?

P(class=-|A=0, B=1, C=0)
=P(A=0, B=1, C=0|class=-)*P(class=-)/P(A=0, B=1, C=0)
= P(A=0|class=-)*P(B=1|class=-)*P(C=0|class=-)*P(class=-)
=

Similarly, P(class=+|A=0, B=1, C=0)
= P(A=0|class=+)*P(B=1|class=+)*P(C=0|class=+)*P(class=+)
=

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Thank you

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